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Indian Institute of Information Technology Raichur

DEEP LEARNING AND MEDICAL IMAGNING

PROJECT REPORT

PROJECT: **Breast Ultrasound Image Classification Using Deep Learning**

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Course: **DEEP LEARNING for MEDICAL IMAGNING**

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1.Introduction.

Breast cancer is one of the most common and life-threatening diseases affecting women worldwide. Early detection plays a critical role in improving treatment success and survival rates. Medical imaging techniques such as mammography, MRI, and ultrasound are widely used for detecting breast abnormalities.

Among these techniques, ultrasound imaging is frequently used because it is non-invasive, affordable, and effective in detecting tumors in dense breast tissue.

However, manual analysis of ultrasound images by radiologists can sometimes lead to diagnostic errors due to human limitations. With the advancement of Artificial Intelligence (AI) and Deep Learning, automated systems can assist healthcare professionals in identifying breast tumors more accurately.

In this project, a Convolutional Neural Network (CNN) is implemented to classify breast ultrasound images into three categories:

- Benign
- Malignant
- Normal

The objective of this project is to develop a deep learning model capable of analyzing ultrasound images and assisting in breast cancer detection.

2. Dataset Description

The dataset used in this project is the Breast Ultrasound Images Dataset (BUSI). The dataset contains ultrasound images categorized into three classes:

- Benign Tumors
- Malignant Tumors
- Normal Breast Tissue

Found 625 images belonging to 3 classes.

Found 155 images belonging to 3 classes.

The dataset is organized into separate folders corresponding to each class, which allows automated loading of images using deep learning frameworks.

During preprocessing, the dataset was split into training and validation sets using an 80% training and 20% validation split.

The dataset loading process resulted in:

- 625 training image
- 155 validation images

3. Methodology

I. 3.1 Data Preprocessing

Before training the neural network, several preprocessing techniques were applied to improve model performance.

Image Rescaling

All image pixel values were normalized by dividing them by 255 so that the pixel range becomes 0–1 instead of 0–255.

Image Resizing

All ultrasound images were resized to 224×224 pixels to maintain a consistent input size for the CNN model.

Data Augmentation

To improve model generalization and prevent overfitting, data augmentation techniques were applied:

- Random rotation
- Zoom transformation
- Horizontal flipping

These transformations increase dataset diversity and allow the model to learn from different image orientations.

II. Convolutional Neural Network Architecture

A Convolutional Neural Network (CNN) was implemented to classify ultrasound images. CNNs are highly effective in image classification tasks because they automatically extract important features.

The architecture of the model includes:

○ Convolution Layers

These layers extract important visual features such as edges, shapes, and tumor textures from ultrasound images.

○ Batch Normalization

Batch normalization helps stabilize training and improve convergence speed.

○ Max Pooling Layers

Max pooling reduces spatial dimensions while retaining the most important features.

- **Flatten Layer**

The flatten layer converts the extracted feature maps into a one-dimensional vector.

- **Dense Layers**

Fully connected layers learn complex relationships between extracted features.

- **Dropout Layer**

Dropout helps prevent overfitting by randomly disabling neurons during training.

- **Output Layer**

The final layer uses Softmax activation to classify the images into:

- Benign
- Malignant
- Normal

Model: 'sequential'

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 222, 222, 32)	896
batch_normalization (Batch Normalization)	(None, 222, 222, 32)	128
max_pooling2d (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_1 (Conv2D)	(None, 109, 109, 64)	18496
batch_normalization_1 (Batch Normalization)	(None, 109, 109, 64)	256
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_2 (Conv2D)	(None, 52, 52, 128)	73856
batch_normalization_2 (Batch Normalization)	(None, 52, 52, 128)	512
...		
Total params: 11,170,243		
Trainable params: 11,169,795		
Non-trainable params: 448		

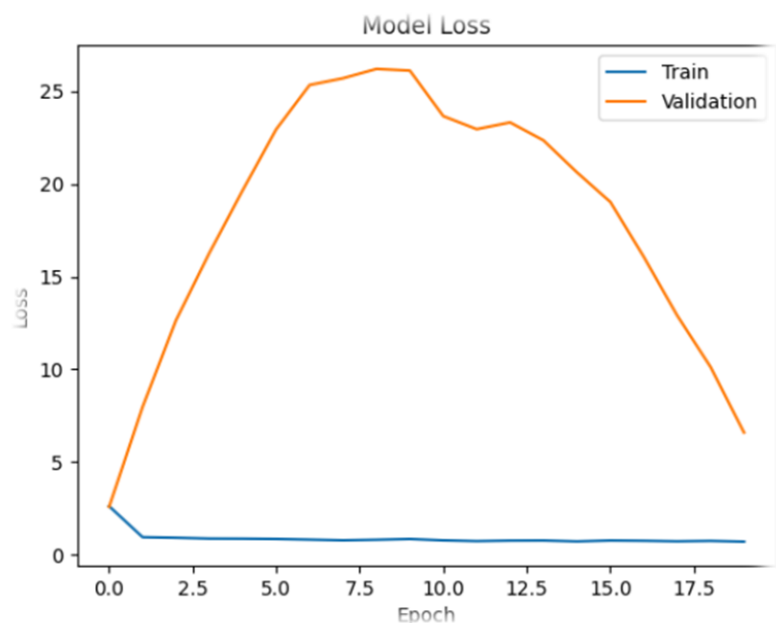
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4. Model Training

The CNN model was trained using the following configuration:

- Optimizer: Adam
- Loss Function: Categorical Cross entropy
- Epochs: 20
- Batch Size: 32

During training, the model learns patterns in ultrasound images and improves its classification performance over multiple epochs.



5. Evaluation Metrics

To evaluate the performance of the CNN model, several metrics were used.

- **Accuracy**

Accuracy measures the proportion of correctly classified images.

Accuracy = Correct Predictions / Total Predictions

- Precision

Precision measures how many predicted positive samples are actually correct.

Precision = $TP / (TP + FP)$

- Recall

Recall measures how many actual positive samples were correctly predicted.

Recall = $TP / (TP + FN)$

- F1 Score

The F1 Score is the harmonic mean of precision and recall.

F1 Score = $2 \times (Precision \times Recall) / (Precision + Recall)$

- Confusion Matrix

A confusion matrix visualizes classification results by comparing predicted labels with actual labels.

It helps identify:

- Correct classifications
- Misclassifications

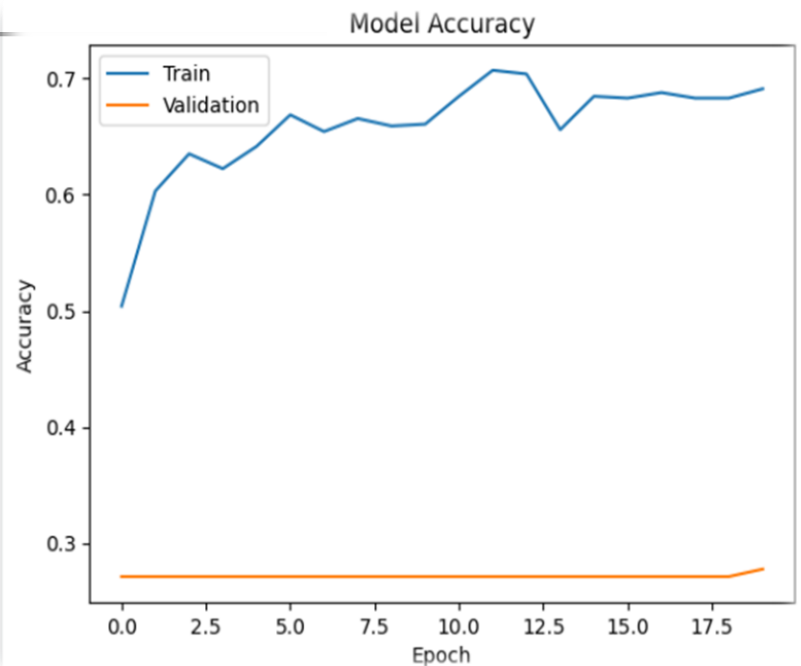
6. Results

The CNN model was trained on the BUSI dataset for 20 epochs. The training process showed improvements in training accuracy while validation performance remained relatively low.

Accuracy Analysis

Training and validation accuracy across epochs.

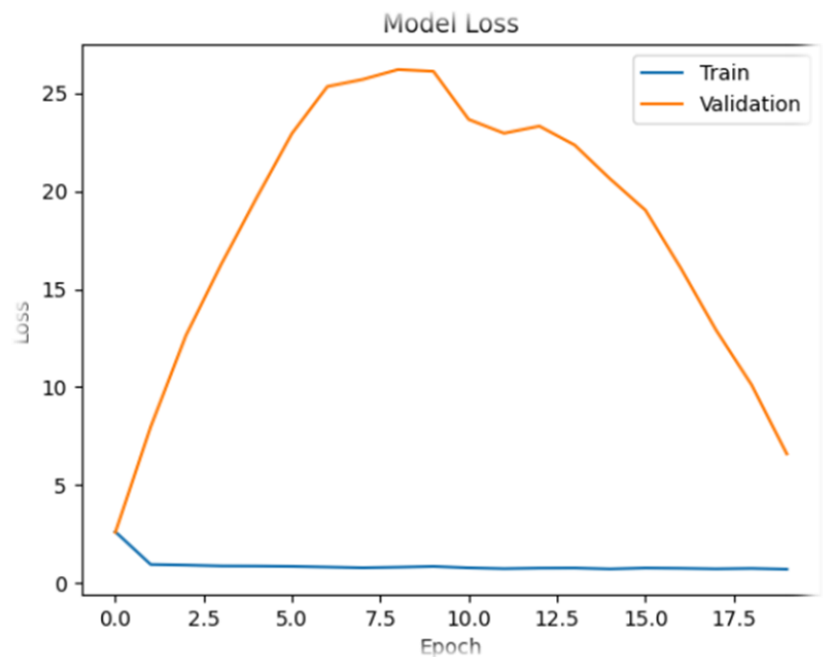
The training accuracy increased gradually and reached approximately 69%. However, validation accuracy remained around 27%, indicating that the model does not generalize well to unseen data.



Loss Analysis

Training and validation loss during training.

The training loss decreased steadily, indicating that the model learned patterns from the training dataset. However, validation loss increased, suggesting overfitting.



Confusion Matrix Analysis

Confusion matrix showing predicted versus actual classes.

The confusion matrix shows that the model predicted many samples as malignant, indicating a bias toward that class.

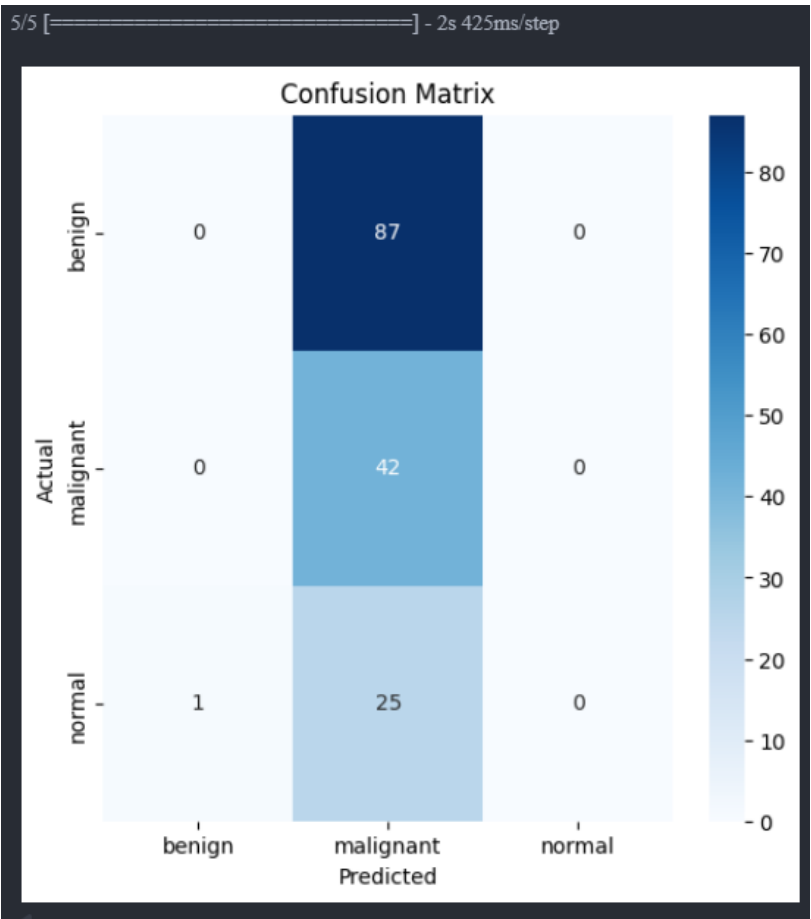
Classification Report

Table 1: Classification report showing precision, recall, and F1-score for each class.

The classification report shows:

- Very low precision for benign and normal classes
- High recall for malignant class
- Overall accuracy approximately 27%

This suggests that the model struggles to differentiate between the three categories.



7. Conclusion

In this project, a Convolutional Neural Network (CNN) was developed to classify breast ultrasound images into benign, malignant, and normal categories using the BUSI dataset.

The model successfully learned patterns from the training dataset and achieved moderate training accuracy. However, the validation accuracy remained low, indicating limitations in model generalization.

Despite these challenges, the project demonstrates the potential of deep learning techniques in assisting medical professionals with automated breast cancer detection.